SC 04 · ROTATIONS AND ATTITUDE

A drone has no potentiometer

Rotations, what an IMU really measures, a filter that blends gyro and accelerometer, and the cascade every flight controller runs.

Rotation matrices, Z-Y-X

$$R = R_z(\psi)\, R_y(\theta)\, R_x(\phi), \qquad v_W = R v_B, \qquad R^{-1} = R^T$$

- columns of R = body axes in world coordinates
- order matters: $R_z R_y \neq R_y R_z$
- **gimbal lock** at $\theta = \pm 90^\circ$: roll and yaw axes coincide

World z up, body x forward / y left / z up -- same as the rotorfault study.

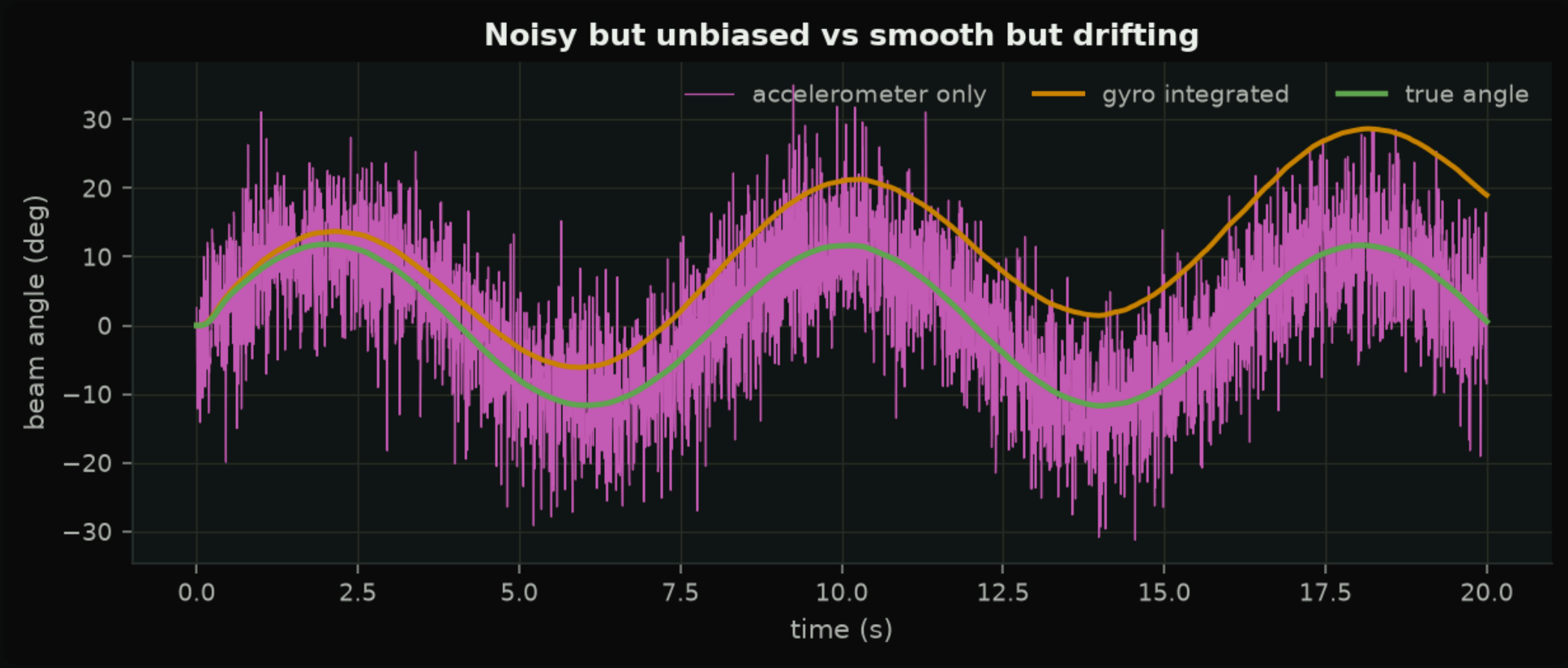
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Quaternions, lightly

$$q = \left(\cos\tfrac{\alpha}{2},\ \sin\tfrac{\alpha}{2}\,\mathbf{n}\right), \qquad v' = q \otimes (0, v) \otimes q^{*}$$

- four numbers, no singularity
- composition = quaternion product
- gyro integration: $q_{k+1} = q_k \otimes \exp(\tfrac{1}{2}\omega\Delta t)$, renormalise

What the IMU says

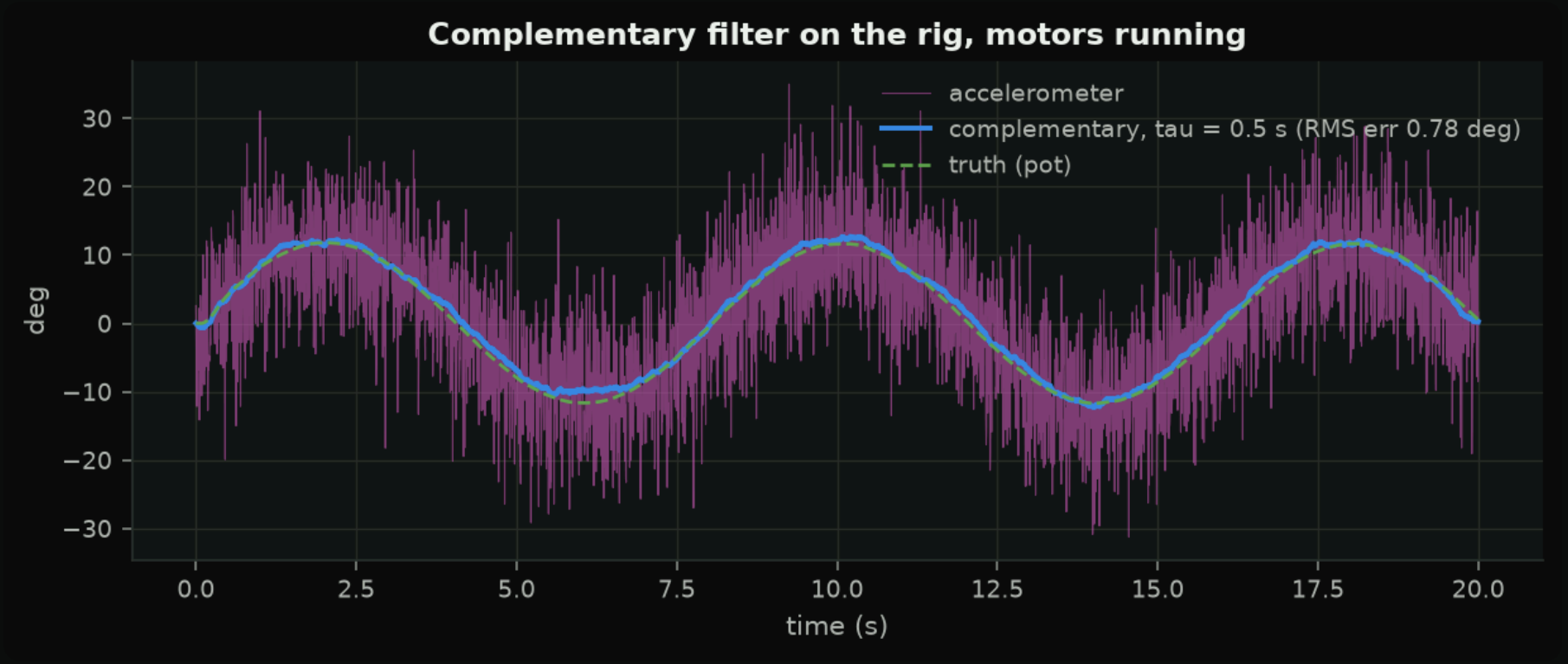


Gyro: smooth, drifts (bias). Accelerometer: unbiased at rest, noisy, fooled by any acceleration.

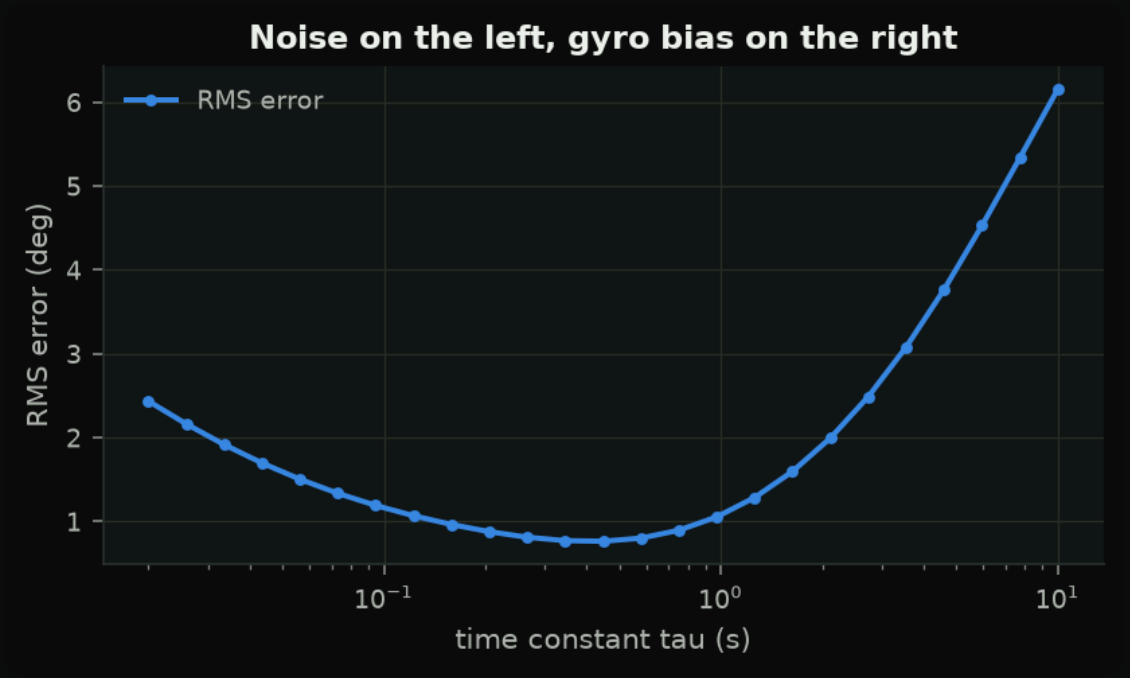
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Complementary filter

$$\hat{\theta}_k = \alpha(\hat{\theta}_{k-1} + \omega_k \Delta t) + (1-\alpha)\theta_{acc}, \qquad \alpha = \frac{\tau}{\tau + \Delta t}$$

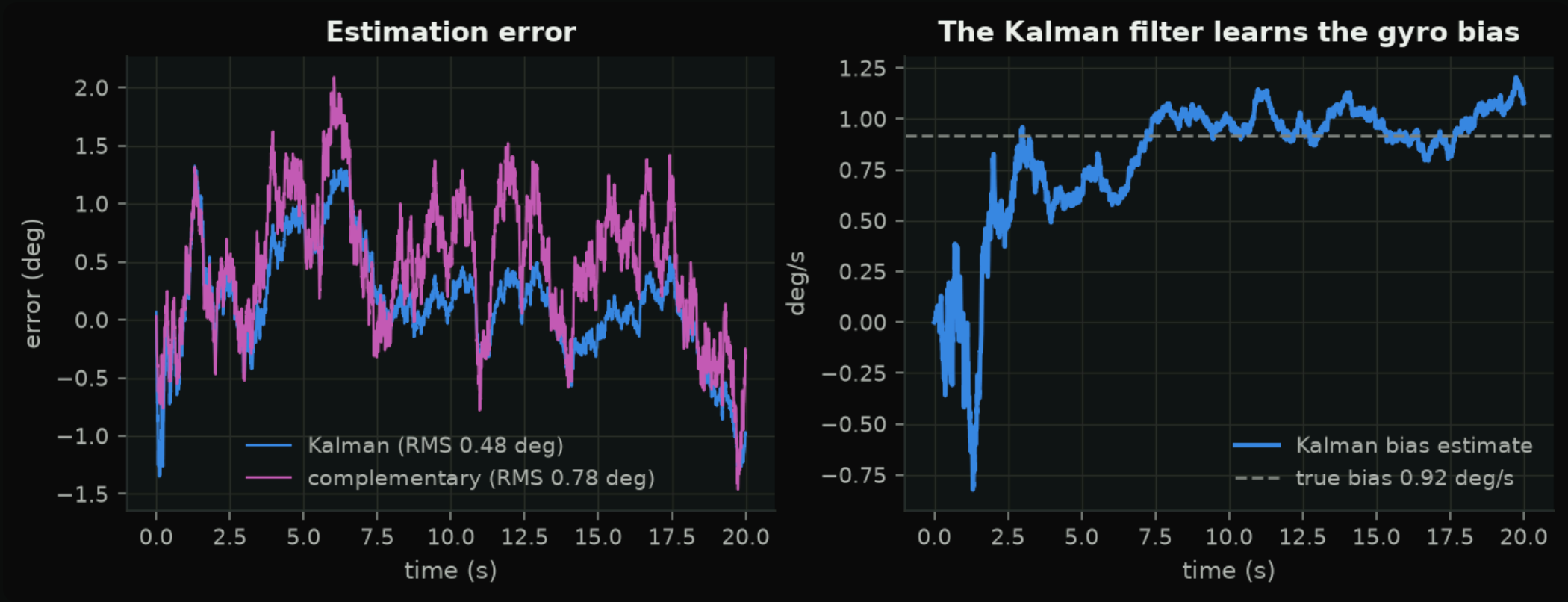


Choosing τ



- small τ: vibration comes through
- large τ: gyro bias leaks in ($\approx b\tau$), slow recovery
- the **Kalman filter** estimates the bias as a state and picks its own gain

Kalman preview: learning the bias



The cascade every flight controller runs

```
angle sp --> [angle loop: P] --> rate sp --> [rate loop: PID] --> mixer --> motors
                  ^ angle estimate                ^ gyro rate
```

- inner loop on the gyro: fast, clean, sees a push immediately
- outer loop 3-10x slower; its output limit is a rate limit
- **tune inside out**

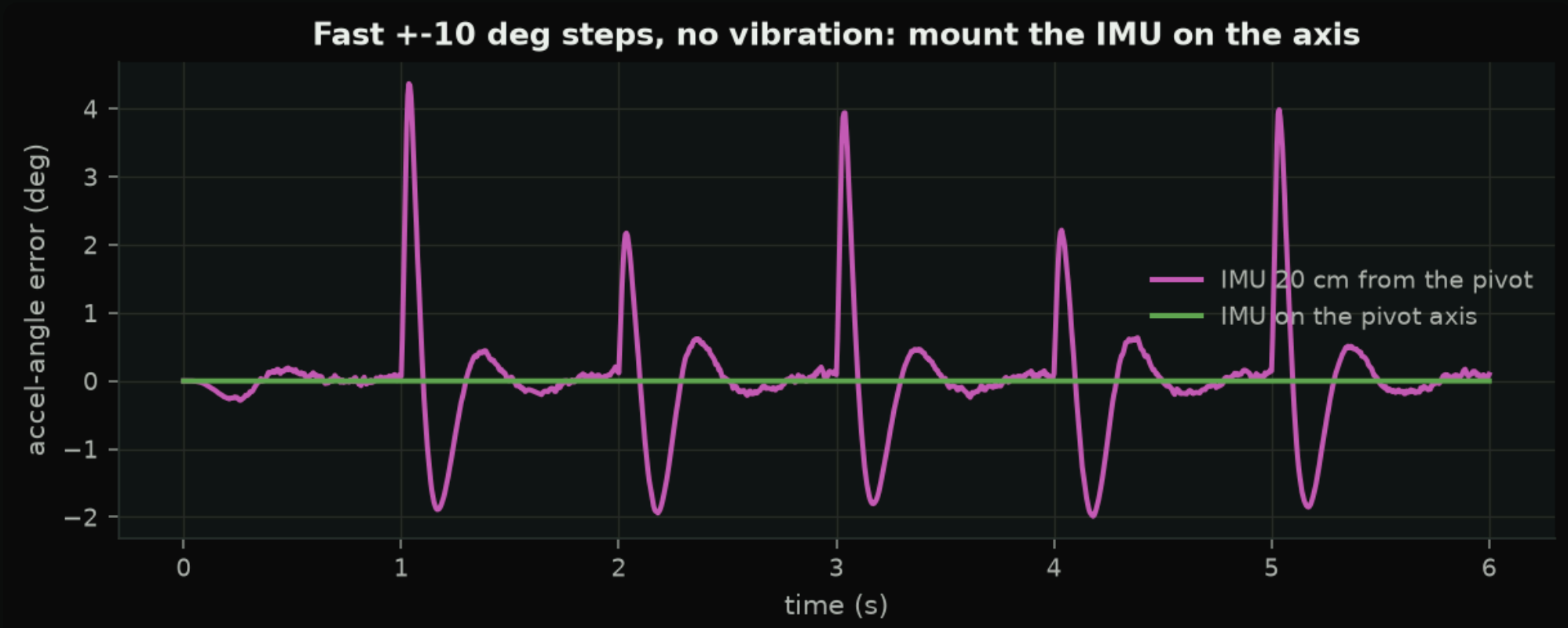
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It rejects disturbances better



Simulation with assumed parameters: tap deviation 7° → 3°. The lab measures yours.

Mount the IMU on the axis



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Lab 4

- **A** `s04_imu_check` : pot vs accelerometer vs gyro vs filter, motors off and on
- **B** `s04_rig_cascaded` : inner loop first, then outer; `kick` for disturbances; `src imu` for drone mode

Check IMU signs by hand **before** arming. Keep the beam still during the gyro calibration.